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Motivation

- Uncertainty quantification (UQ)** is essential for reliable ML, yet methods address either **aleatoric** (data noise) or **epistemic** (limited data/model) uncertainty, not both.
- Conformal methods** (e.g., CQR [1]) and **quantile regression** estimate **aleatoric** but **ignore epistemic**, leading to poor **conditional coverage**.
- Ensemble diversity** (e.g., DE [2], PCS [3]) captures **epistemic** but **ignores aleatoric**. DE [2] combines both but with a fixed ratio ($\lambda=1$).
- CLEAR** adaptively balances *both* via two calibration parameters (γ_1, λ).

Key Idea

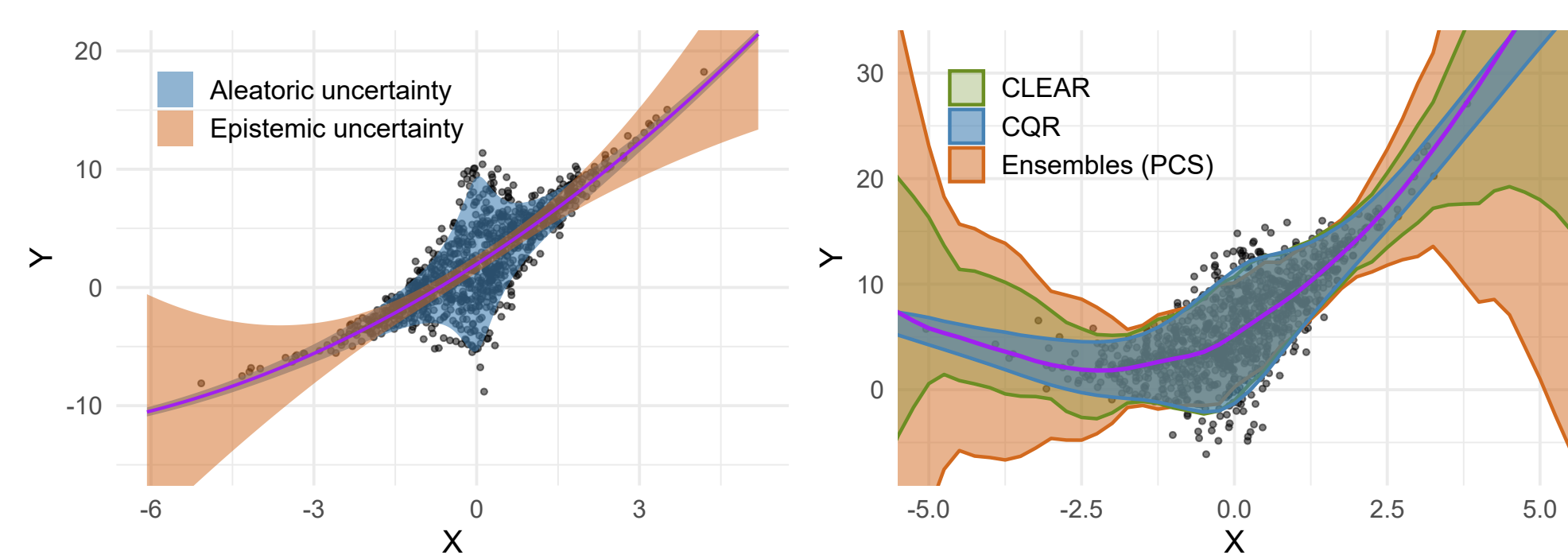


Figure 1: Left: blue = aleatoric, orange = epistemic bands. Right: CQR, PCS, and CLEAR intervals.

$$C(x) = [\hat{f}(x) \pm \left(\frac{\gamma_1}{\text{coverage}} \cdot \text{ale}_{\pm}(x) + \frac{\lambda \gamma_1}{\text{balance}} \cdot \text{epi}_{\pm}(x) \right)]$$

$\hat{f}(x)$: point prediction, γ_1 : ensures **marginal coverage**, λ : **balances** aleatoric vs. epistemic (selected on a held-out set).

- $\lambda = 0$: only aleatoric; $\lambda \rightarrow \infty$: only epistemic
- Adaptive λ** : data-driven $\Rightarrow$ better conditional coverage

Contributions

- Two calibration parameters** (γ_1, λ) to adaptively balance aleatoric and epistemic uncertainty.
- Residual-based quantile regression** for improved aleatoric estimation.
- First to combine PCS** ensemble perturbation intervals with CQR.
- Large-scale benchmarking** on 17 real-world regression datasets.

Why CLEAR?

- No retraining**: wraps around any base model
- Model-agnostic**: works with any estimator pair
- Lightweight**: only 2 parameters (γ_1, λ)
- Principled**: conformal guarantees + proper scoring

Theoretical Guarantees

- Marginal coverage**: true value inside the interval $\geq (1-\alpha)$ of the time, for any finite sample.
- Conditional coverage**: as data grows, coverage holds for each x , not just on average.
- Proper scoring**: quantile loss ensures honest uncertainty estimates.

Key References

- [1] Romano et al. *Conformalized quantile regression*. NeurIPS 2019.
- [2] Lakshminarayanan et al. *Deep ensembles*. NeurIPS 2017.
- [3] Yu & Kumbier. *Veridical data science*. PNAS 2020.
- [4] Agarwal et al. *PCS-UQ*. arXiv 2025.
- [5] Tagasovska & Lopez-Paz. *Single-model uncertainties*. NeurIPS 2019.
- [6] Koenker & Bassett. *Regression quantiles*. Econometrica 1978.

Algorithm

Data is split into training ($\mathcal{D}_{\text{train}}$), calibration ($\mathcal{D}_{\text{cal}}$), and validation ($\mathcal{D}_{\text{val}}$) sets. Target: $(1-\alpha)$ coverage (e.g., $\alpha=0.1$ for 90%).

Step 1: Estimate epistemic uncertainty on $\mathcal{D}_{\text{train}}$

PCS ensembles [3, 4] yield upper/lower epistemic bounds.

Step 2: Estimate aleatoric uncertainty on $\mathcal{D}_{\text{train}}$

Quantile regression on residuals yields upper/lower aleatoric bounds.

Step 3: Calibrate on $\mathcal{D}_{\text{cal}}$ for each candidate λ

Find the smallest γ_1 so the interval covers $\geq (1-\alpha)$ of calibration points (conformal guarantee).

Step 4: Select best λ^* on $\mathcal{D}_{\text{val}}$

Pick λ that minimizes quantile loss [6] $\Rightarrow$ tightest valid intervals.

Simulation Results

We use a synthetic benchmark to test the interval behavior under controlled noise, reporting calibration and sharpness.

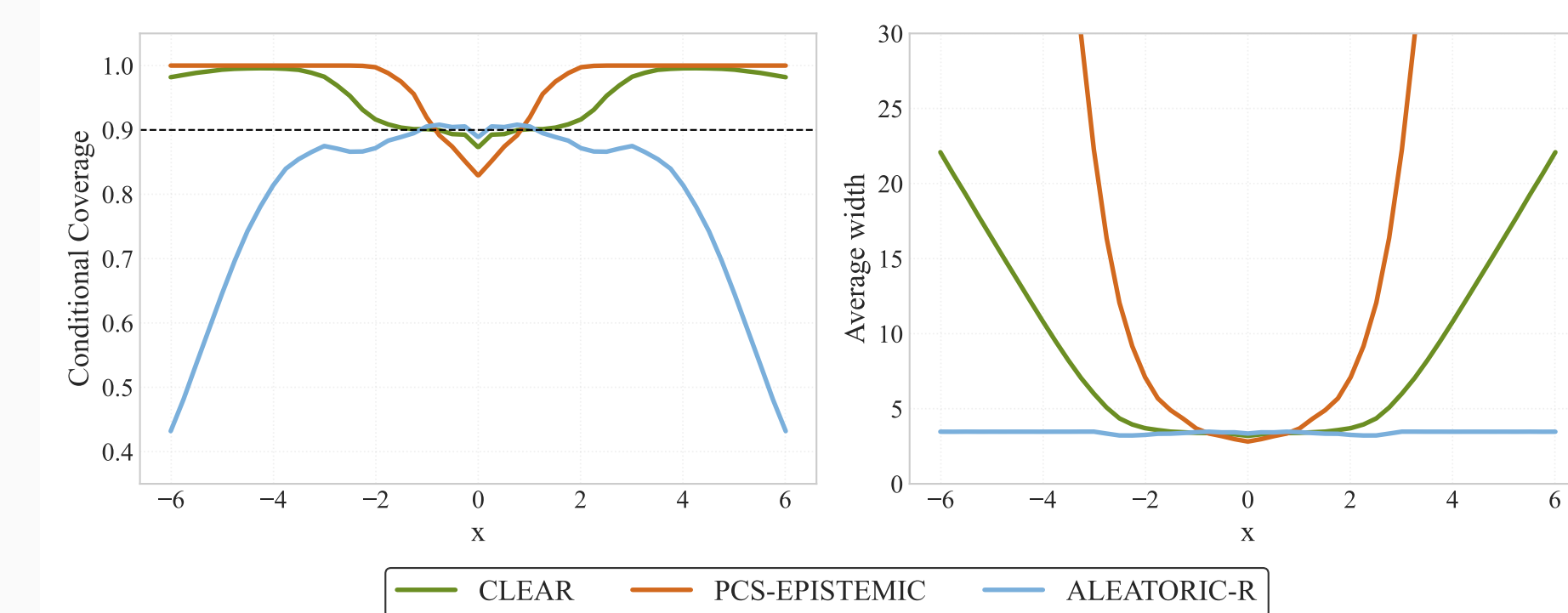


Figure 2: Univariate homoskedastic simulation (100 runs). CLEAR maintains target coverage in extrapolation. Dashed line = 0.9 target.

Real-World Benchmark: 17 Datasets

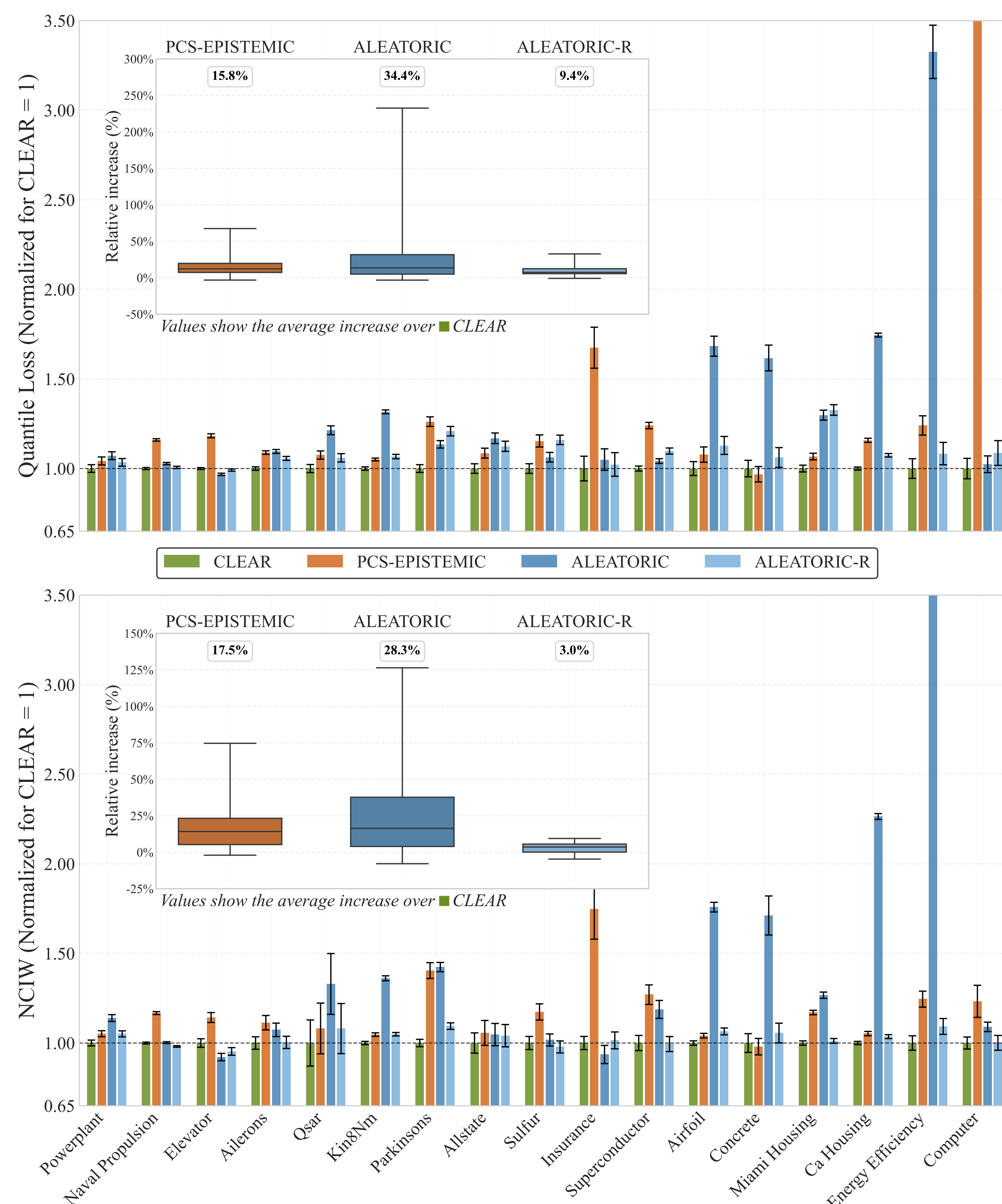


Figure 3: Quantile loss (sharpness) and interval width, normalized to CLEAR (1.0). Lower = better for baselines.

CLEAR's intervals are tighter than individual parts: +28.3% for aleatoric, +17.5% for epistemic

Adaptive combination outperforms either component.

Case Study: Ames Housing

Varying predictors reveals adaptive behavior:

	Method	Width (\$)	Coverage	Quantile loss
2 feat.	PCS	107,880	87%	3,818
	CQR	104,741	90%	3,448
	CLEAR	95,177	89%	3,131
All feat.	PCS	57,594	89%	1,922
	CQR	62,398	88%	2,194
	CLEAR	55,910	88%	1,923

Table 1: Ames Housing (90% coverage target).

- Few feat. $\Rightarrow$ **aleatoric dominates** ($\lambda \approx 0.64$)
- All feat. $\Rightarrow$ **epistemic dominates** ($\lambda \approx 14.5$)
- λ **adapts automatically** to the data regime!

Conclusion

- CLEAR adaptively balances** epistemic & aleatoric uncertainty, outperforming baselines on **17 datasets**.
- Compatible with any pair** of estimators (PCS+CQR, DE+SQR).
- Up to **28.3% tighter prediction intervals** while maintaining valid coverage.
- Theoretical guarantees for **marginal & conditional coverage**.
- Open-source code: `pip install clear-uq`

Generality: Deep Ensembles + SQR

Also works with DE [2] + SQR [5]:

Metric	DE	SQR	DE-c.	SQR-c.
Coverage (PICP)	+0.05	−0.66	−0.09	−0.15
Interval width (NCIW)	+28.6	+13.4	+27.9	+13.2
Sharpness (QL)	+24.0	+13.7	+24.1	+10.1

Table 2: Mean % improvement of CLEAR (95% coverage).

$\Rightarrow$ **CLEAR again improves over its own** aleatoric-only and epistemic-only components when using DEs and SQR.